

Container-based High-Performance Computing Approaches for Simulating Large-Scale Biological Neural Networks

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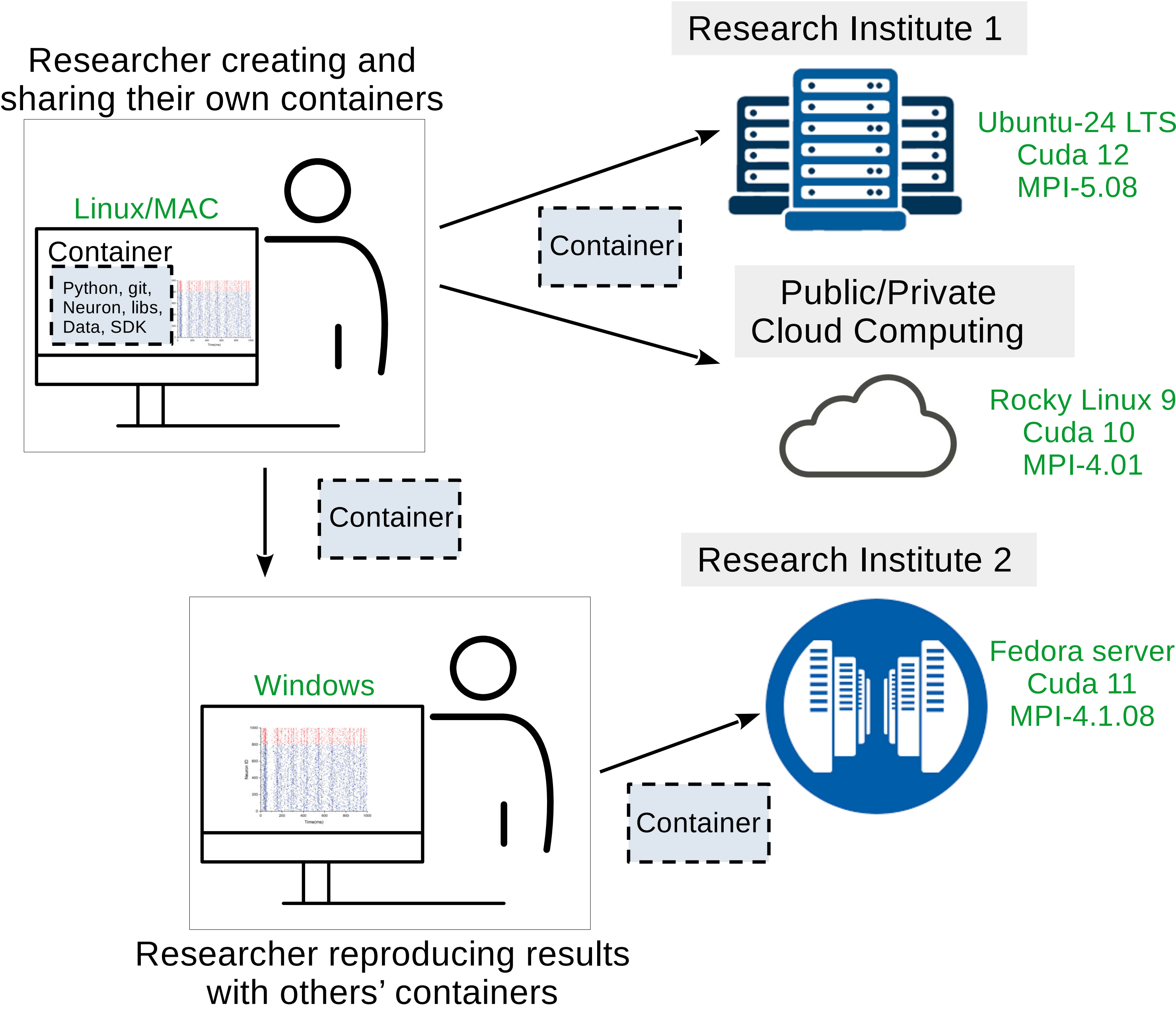
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Introduction

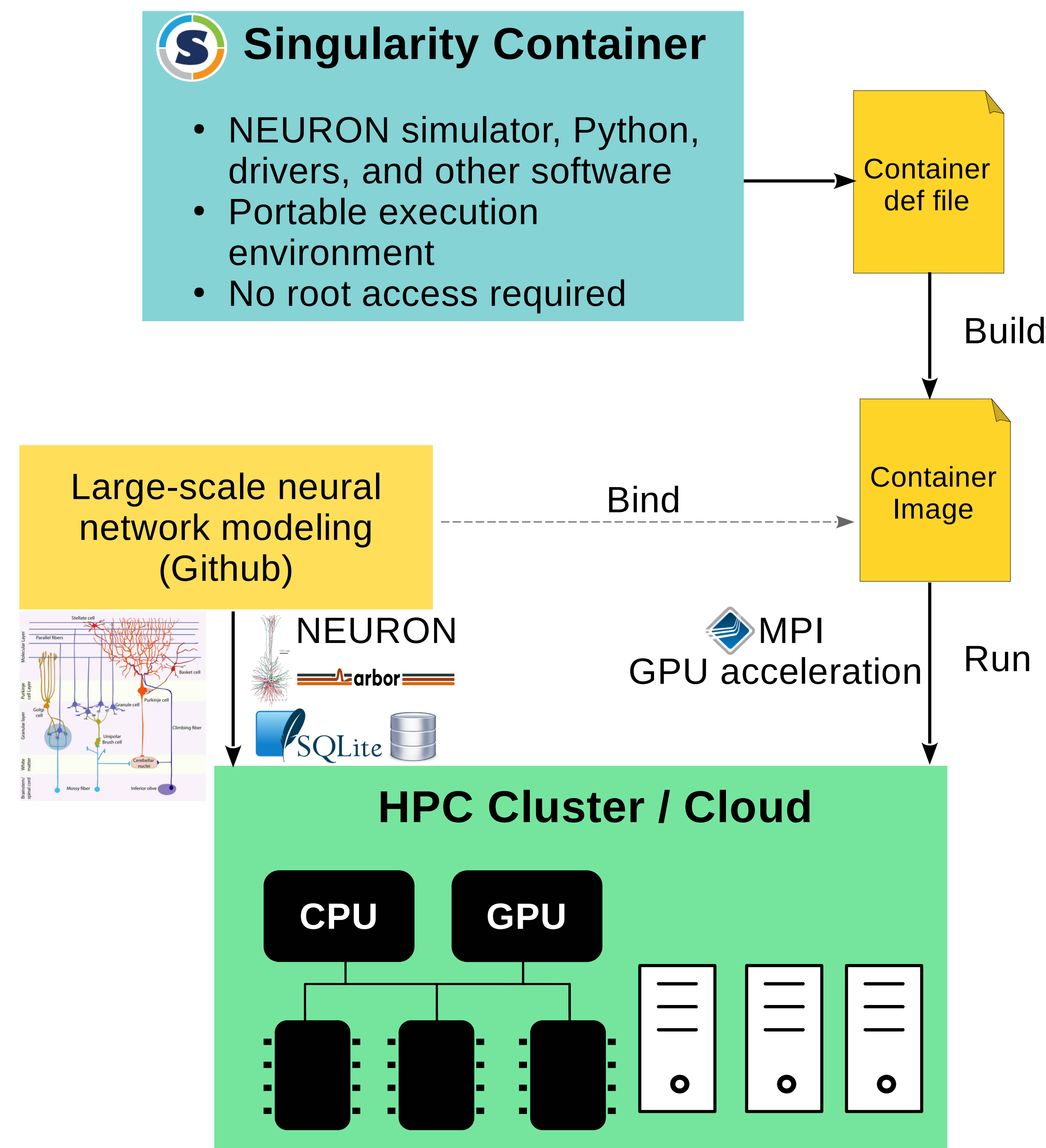
1. Modeling **large-scale, biophysically detailed neural networks** increasingly relies on **high-performance computing (HPC)**.
2. However, deploying simulations of such networks across diverse HPC environments remains challenging due to **software incompatibilities** and **complex dependencies**.
3. We address this issue using **Singularity containers** that encapsulates the entire simulation environment, enabling neuroscientists to achieve portability and reproducibility without requiring root privileges on HPC/Cloud systems.
4. We give practical demonstrations by benchmarking HPC environments with **cerebellar mossy fiber–granule cell network** simulations.

Containerization for Scientific Computing

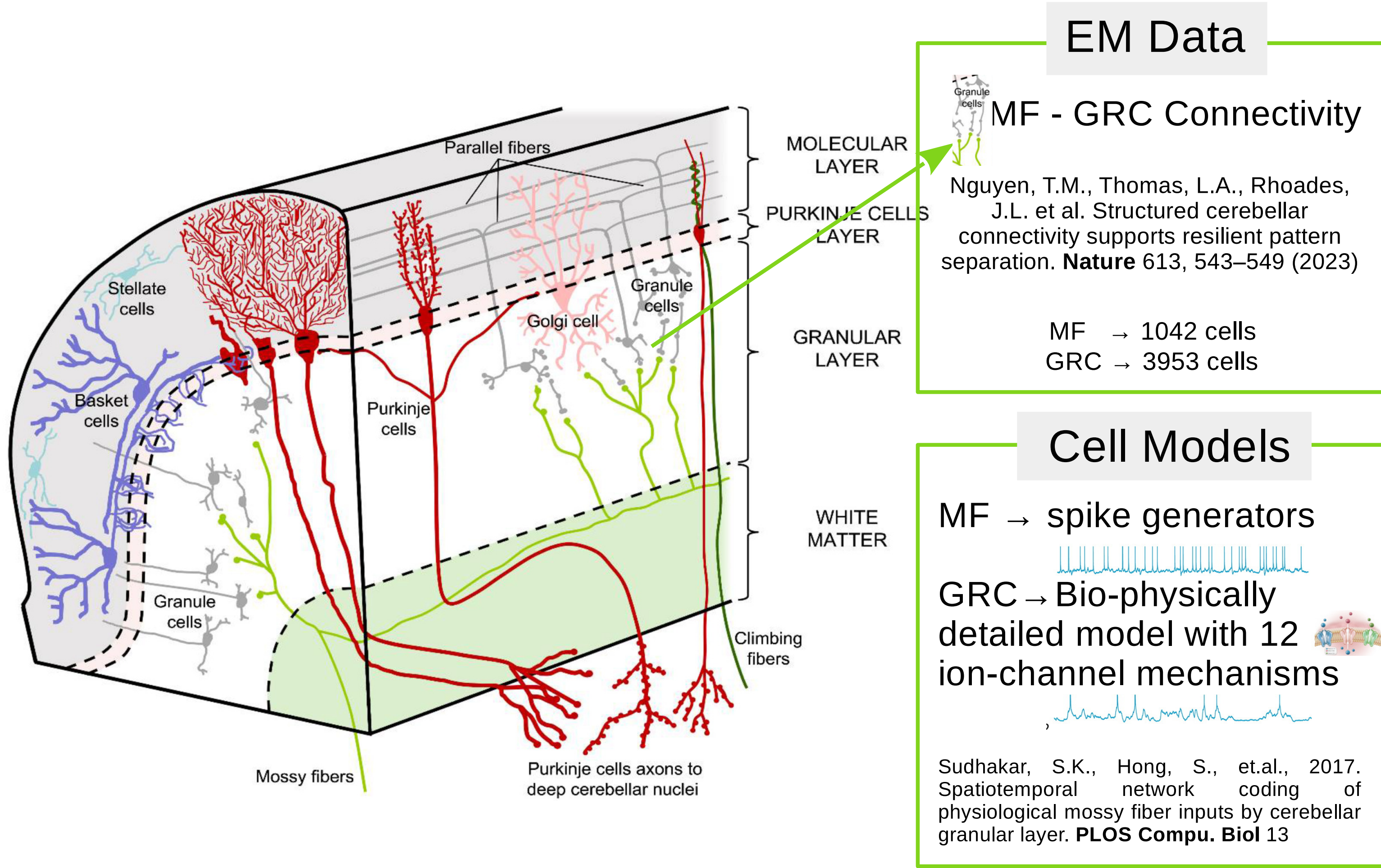
A container is a lightweight, standalone, virtualized executable environment that includes all dependencies required to run large-scale scientific models.



Our Computational Framework



Benchmarking Data and Network: Cerebellar Mossy Fiber → Granule Cell



We run multiple copies ($n = 280$) of the network connectivity model simultaneously to simulate extreme computational workloads.

This allows us to rigorously assess the scalability and performance limits of GPUs under high-demand conditions.

Table: Network Configuration

Network elements	Count (millions)
MF	0.30
GRC	1.09
Compartments	3.29
Synapses	3.47

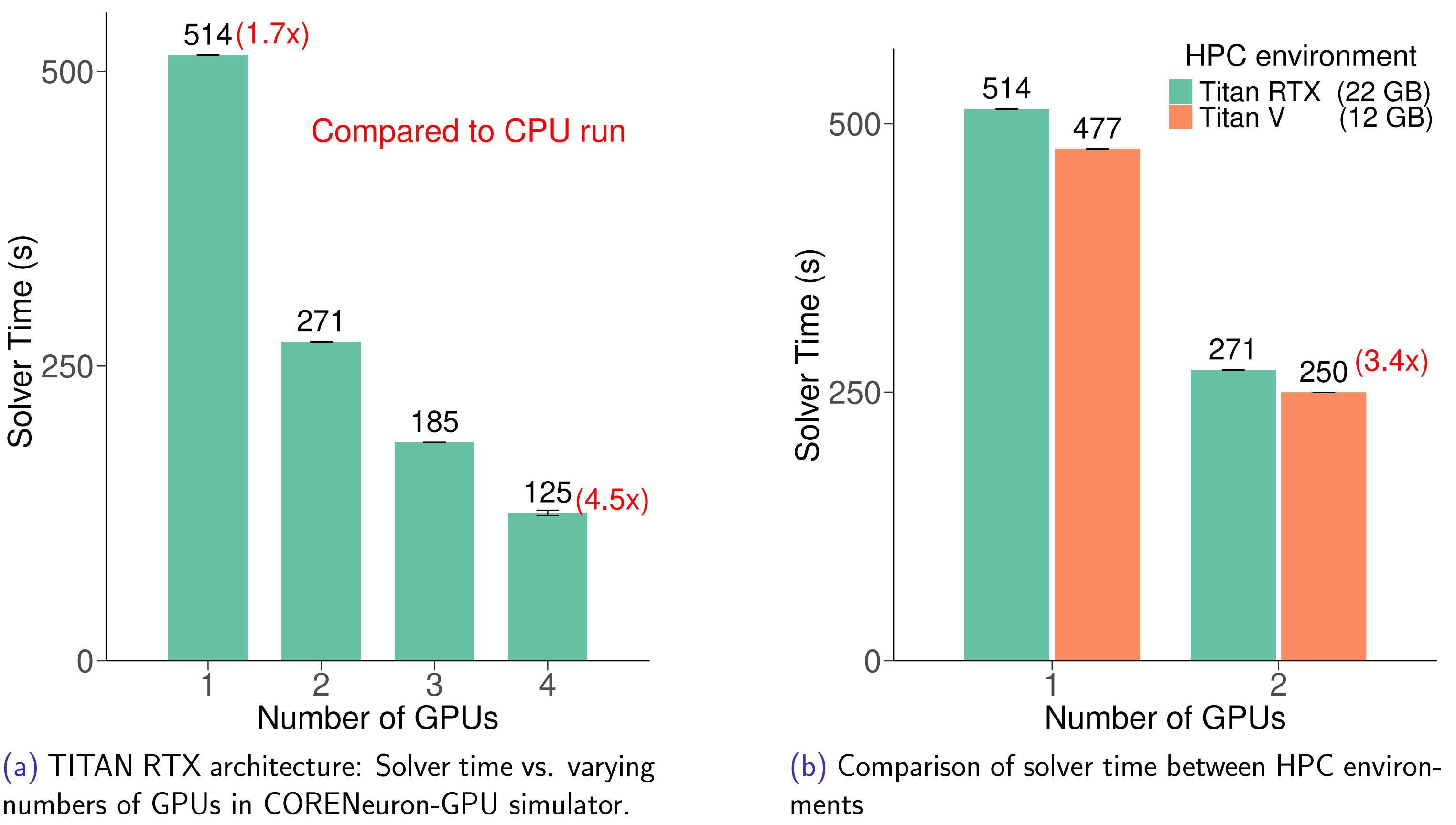
Table: Simulation Setup

Simulation time	1 s
Simulation software	CoreNEURON

Benchmarking HPC Environments

Table: Performance in CPU parallel computing (30-task MPI)

CPU	Core	Solver Time (s)	Simulator
Intel(R)-Xeon	30	564	CoreNEURON-CPU



TITAN RTXs outperform 30-core CPUs by up to 4.5x, with performance improving as GPU count increases.

Future works:

- 🔧 Develop a biologically realistic cerebellar circuit model including granule layer (with Golgi cells), Purkinje layer, and other interneurons.
- 🔧 Benchmark network performance across multiple institutions to evaluate and understand the scalability and performance limits of HPC systems.

Acknowledgment

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